



Emmanuel Amaury Fuentes Venegas

Tech Lead / Senior Full-Stack Engineer

✉ fuentesamaury@hotmail.com ☎ [+52 442-130-47-77](tel:+524421304777) 📍 Mexico, Querétaro 🌐 [Portafolio Web](#)

Summary

Tech Lead and Senior Full-Stack Engineer with 6+ years building and owning a multi-tenant telecom and fintech platform serving ~28 white-label MVNO brands in Mexico. Lead contributor across the entire stack: **#1 committer on the Ruby on Rails core platform (2,094 of 6,649 commits)**, on the Ember admin CRM (1,554 of 5,899), and on both Kotlin mobile applications.

Deep expertise in multi-tenant architecture, payment systems (10 gateways unified behind a single protocol), carrier-grade telecom integrations, hands-on AWS cloud infrastructure, and cross-platform mobile — including leading the migration to Kotlin Multiplatform. Comfortable owning a system end to end: AWS infrastructure, schema, API, background pipelines, mobile clients, and the CI/CD that ships it.

Education

Instituto Tecnológico de Querétaro (ITQ)
Computer System Engineer

Bachelor of Engineering
Querétaro, Qro • August 2016 – January 2022

Languages

Spanish
Native



English
Professional



Profiles

LinkedIn
[amaurylink](#)

Github
[saptial4](#)

Skills

Backend & APIs

Expert
Ruby on Rails, PostgreSQL, Sidekiq / Redis, REST
API Design, RSpec



Frontend

Good
Ember.js, JavaScript, TypeScript, Node.js, SCSS



DevOps & Delivery

Expert
Docker, GitLab CI/CD, Capistrano, HashiCorp Vault,
CloudWatch & OpenTelemetry



Mobile

Expert
Kotlin Multiplatform, Compose Multiplatform,
Kotlin / Android SDK, Swift, fastlane



AWS Cloud

Expert
EC2 & Auto Scaling, S3 & CloudFront, RDS
(PostgreSQL), ELB & Route 53, VPC, IAM & ACM



Architecture & Practice

Expert
Multi-Tenant SaaS, Clean Architecture, Payment
Integrations, Technical Leadership



Koonol México

Tech Lead / Senior Full-Stack Engineer

Zapopan, Jal. (Remote)

April 2021 – April 2026

- **Platform Ownership:** Owned the core MVNO platform — a multi-tenant Ruby on Rails API (108 tables, 203 models, 371 REST endpoints, ~95K LOC production / ~151K LOC tests) powering ~28 white-label telecom brands. Largest contributor with 2,094 of 6,649 commits. Architected multi-tenancy as a tenant tree with encrypted per-tenant carrier credentials and cascading theme settings — one deployment, every brand.
- **Payment Systems:** Unified 10 payment gateways (Openpay, Stripe, Conekta/Digital FEMSA, MercadoPago, Kushki, Clip, Toku, DPCard, DPVale, Beel) plus OXXO cash references behind a single process → commit charge protocol with per-gateway error capture, making each new processor a contained change.
- **Carrier Integrations:** Built the Altán Redes provisioning integration (~40 operations: activation, suspension, IMEI lock, number portability, HSS subscriber profiles), SFTP batch ingestion of CDR/mobility/NIR reports, a second MVNE (OXIO), Mexico's number-portability clearinghouse and CFDI/SAT e-invoicing — 18 vendor clients standardised on one contract.
- **Reliability & Correctness:** Engineered a 26-reason retailer balance ledger with pessimistic row locking, an anti-fraud rate limiter with six configurable rules, a retry library with exponential backoff and bounded webhook redelivery, and concurrency-capped job queues (95 jobs / 18 queues / 24 schedules) tuned to carrier API limits.
- **AWS Cloud Infrastructure:** Designed and operated the AWS infrastructure behind ~28 white-label brands and 100+ platform deployments (CRMs, CMS dashboards, public top-up portals and APIs). Architected for redundancy and high availability across multiple Availability Zones — EC2 with Auto Scaling Groups behind Elastic Load Balancing for horizontal and vertical scaling, Multi-AZ RDS (PostgreSQL), S3 static hosting fronted by CloudFront for the public pages, Route 53 DNS with health-check failover, ACM-managed TLS, VPC isolation and CloudWatch monitoring — standing up new client environments as day-to-day work.
- **Mobile Leadership:** Led the Kotlin Multiplatform migration, authoring 63% of the codebase (242 of 386 commits): Clean Architecture + MVVM with 54 use cases, 43 screens and 30 ViewModels shared across Android and iOS, with Kotlin/Native↔Swift interop and runtime theming. Automated white-label releases via fastlane to Play Store and TestFlight, with per-brand secrets injected from HashiCorp Vault at build time.
- **Standards & Mentorship:** Established GitLab CI with RSpec (432 specs), RuboCop, Brakeman/Snyk SAST, 42Crunch, an 8,423-line OpenAPI spec and OpenTelemetry tracing across 146 releases. Mentored engineers through code review, training and onboarding; led architecture decisions across backend, mobile and frontend.

Gurúcomm

Software Developer

Querétaro, Qro

January 2020 – April 2021

- Grew into a core contributor on the Ruby on Rails API and Ember.js CRM that became the company's flagship telecom platform, taking ownership of subscriber lifecycle, billing and dealer-network features.
- Built and maintained the retailer/dealer Android sales application in Kotlin (MVVM, Room), which went on to ship across 17 white-label brands.

Gurúcomm

Trainee Developer

Querétaro, Qro

October 2019 – December 2019

- Joined the team building corporate telecom platforms with Ruby on Rails and Ember.js; delivered first production contributions to the customer-facing self-service portal.

Multi-Tenant MVNO Core Platform

Period: 2020 – 2026

Role: Tech Lead / Lead Backend Engineer | **Stack:** Ruby on Rails, PostgreSQL, Sidekiq, Redis, AWS

The business-support system behind ~28 white-label mobile carriers: subscriber lifecycle, SIM provisioning, number portability, billing cycles, dealer network and payments. 108 tables, 371 endpoints, 102 authorization policies, and a subscriber model coordinating three orthogonal state machines. Multi-tenancy as a tenant tree with encrypted per-tenant carrier credentials. **Largest contributor: 2,094 of 6,649 commits.**

Unified Multi-Gateway Payment Platform

Period: 2021 – 2026

Role: Lead Engineer | **Stack:** Openpay, Stripe, Conekta/FEMSA, MercadoPago, Kushki, Clip, Toku, DPCard, DPVale, Beel

Consolidated ten payment processors plus OXXO cash references behind one charge protocol with per-gateway error persistence and provider-specific webhook handling. Paired with a 26-reason double-entry-style dealer balance ledger using pessimistic locking for concurrency safety, a configurable anti-fraud rate limiter, and CFDI/SAT e-invoicing for Mexican tax compliance.

Kotlin Multiplatform Consumer App

Period: 2024 – 2026

Role: Mobile Architect / Lead Engineer | **Stack:** Kotlin Multiplatform, Compose Multiplatform, Ktor, Koin, Room

Led the migration of the consumer mobile ecosystem to Kotlin Multiplatform, replacing separate native codebases with one shared commonMain (336 files) targeting Android and iOS. Clean Architecture with 54 use cases, 14 repositories, 43 screens and 30 ViewModels; Swift interoper via swiftklib and cinterop; server-driven runtime theming across 6 brands. **Authored 63% of the codebase (242 of 386 commits).**

Multi-Client AWS Cloud Infrastructure

Period: 2020 – 2026

Role: Cloud / DevOps Engineer | **Stack:** AWS — EC2, Auto Scaling, ELB, S3, CloudFront, RDS, Route 53, ACM, VPC, IAM, CloudWatch

Architected, deployed and operated the AWS footprint for ~28 white-label brands — 100+ platform deployments spanning admin CRMs, CMS dashboards, public top-up portals and REST APIs. Built high-availability environments across multiple Availability Zones: EC2 Auto Scaling Groups behind Elastic Load Balancers for horizontal and vertical scaling, Multi-AZ RDS (PostgreSQL) for database redundancy, and S3 static hosting fronted by CloudFront for fast, resilient public pages. Handled DNS and failover with Route 53, TLS with ACM, network isolation with VPC and security groups, access control with IAM, and monitoring/alerting with CloudWatch — provisioning new client environments as routine work, with availability, redundancy and performance optimization as first principles.

Retailer & Dealer POS Android Application

Period: 2021 – 2026

Role: Lead Android Engineer | **Stack:** Kotlin, MVVM, Room, ZXing

Point-of-sale application for the dealer network: SIM activation, customer signup, number portability, IMEI verification and balance/offering purchase. **17 product flavors** from one codebase, six payment processors integrated natively, and ZXing barcode scanning for SIM/ICCID/IMEI capture. **Top contributor: 232 of 559 commits.**

Telecom Admin CRM

Period: 2019 – 2026

Role: Lead Frontend Engineer | **Stack:** Ember.js, JavaScript, SCSS, AWS S3 + CloudFront

The internal operations CRM: 118 models, 337 routes and 187 components, with shared CRUD base classes, a bitmask-driven permission system gating features per role, and dashboards with chart and PDF export. Deployed per tenant via GitLab CI to S3 + CloudFront with Vault-injected configuration. **Top contributor: 1,554 of 5,899 commits over 6.5 years.**

DEMO Multi-tenant CRM Platform

Period: April 2026 – Present

Role: Fullstack Developer | **Stack:** Ember.js, JavaScript, Docker, Vercel

A multi-tenant CRM frontend for enterprise sales management, built around modular component architecture and integration with complex backend APIs across diverse tenant requirements.

<https://crm-demo.soyamaury.dev/>